

# FUSIONCORE $v_0$

## Phase 4 & 5 — Benchmarking, Forensic Audit, Comparative Evaluation

*Notebook 04x, 05x | Technical Deep-Dive Analysis*

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## 1. Abstract

This report presents a rigorous technical analysis of the SOTA benchmarking, forensic audit, and comparative evaluation pipeline implemented in FusionCore  $v_0$ , Phases 4 and 5 (Notebooks 04a–04d and 05a–05e). The pipeline operates on the unified, regime-normalised, physics-informed feature dataset FD00u produced by Phase 3, comprising 129,331 internal training rows and 31,028 validation rows

across 567 training engines, with 104,897 official test rows across 707 held-out engines, drawn from the NASA C-MAPSS FD001–FD004 corpus Saxena & Goebel (2008) [1]. Phase 4 addresses five principal activities:

1. programmatic resolution of all nine transfer items carried forward from Phase 3,
2. XGBoost baseline training and SHAP-based feature attribution on the 91-feature physics-informed manifold,
3. forensic audit comprising a target null test and a physical perturbation reactivity test,
4. disqualification of DeepAR on two architectural grounds and,
5. training of the Temporal Fusion Transformer (TFT) and N-HITS under capacity-scaled, literature-standard hyperparameters without Optuna optimisation.

Phase 5 applies the Iron Wall Protocol to the official test set, producing a four-quadrant evaluation framework encompassing point estimation accuracy (RMSE, MAE), the NASA asymmetric safety score, Critical-band operational classification ( $F_2$  score,  $\beta = 2$ ), and model explainability status.

XGBoost achieves RMSE = 14.85 cycles, MAE = 10.34 cycles, NASA Score = 4,336.3, and Critical-band  $F_2 = 0.9339$  on the 707-engine official test set. TFT and N-HITS produce a NASA score of 10,061,188.1, a ratio of 2,320:1 relative to XGBoost, and a Critical-band recall of 4.5%, rendering them operationally non-viable in their current  $v_0$  configurations. A score collision anomaly — whereby TFT and N-HITS produce identical metrics to four decimal places across both evaluation stages — is identified, diagnosed, and flagged as a blocking gap item for FusionCore  $v_1$  resolution. DeepAR is formally disqualified: it cannot ingest exogenous sensor features and cannot exclude past target values, making it structurally incompatible with the zero-leakage evaluation protocol. A financial impact analysis projects net annual savings of \$13,818,171 for a 100-aircraft fleet under XGBoost deployment at baseline MRO cost assumptions, with positive net savings confirmed across the full conservative-to-aggressive literature-validated parameter range. Six programmatic phase gates are applied as acceptance thresholds. All six gates pass on the executed run. Critical findings, performance metrics,

identified engineering gaps, and steering recommendations for FusionCore  $v_1$  — including the rationale for the PiNet physics-informed hybrid architecture — are documented herein.

## 2. Introduction & Scope

This document constitutes the unified technical analysis for Phases 4 and 5 of the FusionCore  $v_0$  pipeline. Phase 4 encompasses the XGBoost baseline training, forensic audit, and SOTA benchmarking against three candidate neural architectures: Temporal Fusion Transformer (TFT), N-HiTS, and DeepAR. Phase 5 applies the Iron Wall Protocol to the NASA C-MAPSS official test set, producing a four-quadrant comparative evaluation across all qualified models.

The primary objectives of this report are:

- (i) to document all Phase 3 transfer item resolutions,
- (ii) to characterise the XGBoost baseline and SOTA model performance against aeronautical safety-relevant metrics,
- (iii) to quantify the operational and financial implications of model selection for a commercial airline maintenance operator and,
- (iv) to provide a formal steering brief for FusionCore  $v_1$ , including identification of outstanding gaps, preprocessing continuity, and the PiNet hybrid architecture rationale.

All feature matrices are derived from the 709-engine NASA C-MAPSS training corpus Saxena & Goebel (2008) [1], split 80/20 by engine into internal train and validation sets (567/142 engines). The full test corpus comprises 707 engines. The feature manifold contains 91 engineered features for XGBoost and 87 features for neural architectures (four `s16`-family features excluded following SHAP ablation). All metrics are computed at the final observed cycle per engine, consistent with the C-MAPSS evaluation convention of [1].

## 3. Phase 4 — Benchmarking & Forensic Audit

### 3.1 Transfer Item Resolution

Phase 3 concluded with nine deferred transfer items requiring resolution prior to model training. Items classified MANDATORY constitute gate-blocking prerequisites; items classified ADVISORY are documentation obligations. All nine items were addressed in notebook 04a before any model received training data. Table 1 presents the complete disposition record.

**Table 1 — Transfer Item Resolution Register (Phase 3 → Phase 4)<sup>1</sup>**

| Item | Priority  | Description                                             | Disposition                                                                               |
|------|-----------|---------------------------------------------------------|-------------------------------------------------------------------------------------------|
| T1   | ADVISORY  | G3 class imbalance — terminal bias assessment           | CONDITIONAL RESOLVED — no bias detected (Critical band mean residual = 2.33 cycles)       |
| T2   | MANDATORY | <code>s16</code> family ablation via SHAP               | RESOLVED — <code>s16</code> family = 0.88% total SHAP: excluded from NN matrices          |
| T3   | MANDATORY | <code>s9_s14_ratio</code> — SHAP audit + [P1, P99] clip | RESOLVED — clipped to [−12.62, 14.74]; post-clip $\sigma = 2.71$ vs raw $\sigma = 650.60$ |
| T4   | ADVISORY  | <code>s8_s13_ratio</code> — SHAP audit                  | DOCUMENTED — SHAP = 0.077 (rank 51/91); adds beyond raw pair, retained                    |

<sup>1</sup> Appendix D — Transfer Item Resolution Register gives a more in-depth coverage of Table 1.

| Item | Priority  | Description                                                | Disposition                                                                                                         |
|------|-----------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| T5   | MANDATORY | Yeo-Johnson on CPR & $E_{thermal}$ for NN inputs           | RESOLVED — <code>log1p + StandardScaler</code> cascade; skewness: CPR 41.20 → -0.028, $E_{thermal}$ 171.82 → -0.011 |
| T6   | MANDATORY | Inverse-frequency regime weighting for NNs                 | RESOLVED — regime 0 (42.4%) weight = 0.305; minority regimes weighted $\approx 1.20$                                |
| T7   | ADVISORY  | CPR denominator guard (denominator = <code>s3</code> )     | DEFERRED to $v_1$ — z-space CPR mitigated by T5; raw data dependency not justified for $v_0$                        |
| T8   | ADVISORY  | <code>fillna(0)</code> convention at trajectory boundaries | DOCUMENTED — cycle-1 rows = 0.44% of data; zero-fill is physically justified                                        |
| T9   | ADVISORY  | Phase 3 Gate 2 reporting discrepancy                       | DOCUMENTED — display artefact only; feature count 91 confirmed correct                                              |

## 3.2 XGBoost Baseline Training

The XGBoost baseline was trained on the internal training split (`x_train`, 129,331 rows × 91 features) with early stopping evaluated against the validation split (`x_val`, 31,028 rows × 91 features). The model was configured according to the specification of Chen & Guestrin (2016) [2]: `n_estimators=1000`, `learning_rate=0.05`, `max_depth=6`, `subsample=0.8`, `colsample_bytree=0.8`, with early stopping patience of 50 rounds. Table 2 presents the training and validation results.

**Table 2 — XGBoost Baseline Training & Validation Metrics (FD00u Combined)**

| Metric                | Training Set | Validation Set |
|-----------------------|--------------|----------------|
| RMSE (cycles)         | 9.6787       | 14.9896        |
| MAE (cycles)          | 6.4733       | 9.9627         |
| NASA Asymmetric Score | 213,795.6    | 176,641.2      |

The RMSE and MAE metrics are defined over the  $N$  final-cycle predictions per engine:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

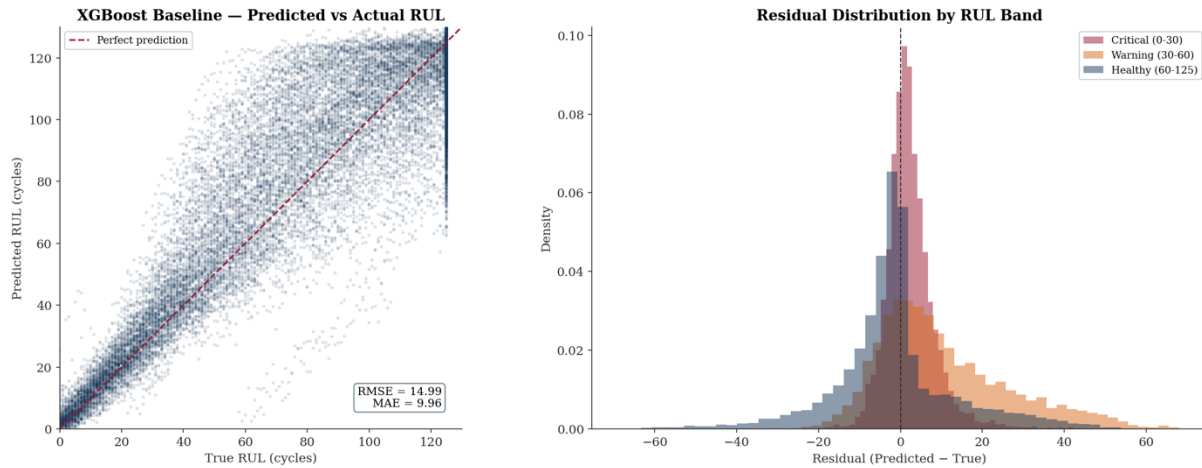
where  $y_i$  denotes the true Remaining Useful Life for engine  $i$  and  $\hat{y}_i$  the model prediction, with  $N = 142$  on the validation set and  $N = 707$  on the test set.

Figure 1: The left panel presents the predicted versus actual RUL scatter across all validation cycles. The concentration of points along the perfect-prediction diagonal confirms systematic tracking of the degradation trajectory, with dispersion increasing at the RUL cap boundary of 125 cycles — an expected artefact of the piecewise-linear label ceiling, where all early-life cycles are assigned an identical target value and the model correctly learns to saturate predictions at the cap. The vertical density streak at true RUL = 125 reflects this label compression rather than any prediction pathology.

The right panel decomposes the signed residual distribution ( $\hat{y}_i - y_i$ ) by operational band and is the more diagnostically informative of the two plots. All three bands — Critical (0–30 cycles), Warning (30–

60 cycles), and Healthy (60–125 cycles) — are centred close to zero, confirming the absence of systematic directional bias across the RUL range. The Critical band (red) exhibits the narrowest and tallest distribution, indicating that the model is most tightly calibrated precisely where operational consequence is highest. The Warning and Healthy bands show progressively heavier right tails, reflecting a moderate tendency toward late prediction at longer RUL horizons — consistent with the label-ceiling saturation effect and consistent with the T1 finding of  $\bar{\epsilon}_{\text{crit}} = +2.33$  cycles. Critically, no band exhibits a heavy left tail, confirming that large early predictions — the operationally safer error direction — dominate the outlier population over large late predictions.

**Figure 1 — XGBoost Baseline: Predicted vs Actual RUL and Residual Distribution by Operational Band (Validation Set, N = 142 engines)**



### 3.2.1 What the Numbers Actually Say

Table 2 presents three metrics across two evaluation splits for the XGBoost baseline trained on the unified FD00u dataset. Each metric quantifies a different aspect of predictive accuracy, and the relationship between them — particularly the apparent inversion between symmetric and asymmetric scores across splits — carries specific diagnostic significance that is explained in full below.

#### A. RMSE and MAE — The Symmetric Error Metrics

Both RMSE and MAE are computed over the  $N$  final-cycle predictions per engine: the model predicts RUL at the last observed cycle of each engine trajectory, and the signed error for engine  $i$  is:

$$e_i = \hat{y}_i - y_i$$

- RMSE = 14.99 cycles (validation). The model's predictions deviate from true RUL by approximately 15 engine cycles in the root-mean-squared sense. Because RMSE squares errors before averaging, it disproportionately penalises large individual errors: a single engine predicted at 50 cycles remaining when true RUL = 5 contributes 2,025 to the sum of squares, versus a contribution of 16 from an engine predicted 4 cycles in error.
- MAE = 9.96 cycles (validation). The gap between RMSE (14.99) and MAE (9.96) is diagnostically important. The ratio  $\text{RMSE}/\text{MAE} \approx 1.5$  indicates the presence of a tail of larger errors that RMSE detects but MAE smooths over. These tail errors — engines where the model is substantially wrong — are precisely the safety-critical cases that the NASA asymmetric score is designed to amplify.

The generalisation gap between training and validation performance is computed as:

$$\Delta_{\text{RMSE}} = 14.9896 - 9.6787 = 5.31 \text{ cycles } (+54.9\%)$$

$$\Delta_{\text{MAE}} = 9.9627 - 6.4733 = 3.49 \text{ cycles } (+53.9\%)$$

A gap of approximately 54% in both metrics is consistent with moderate, expected overfitting for a  $v_0$  baseline operating under a fixed hyperparameter set without Optuna HPO. The validation RMSE of 14.99 cycles aligns with published SOTA benchmarks for XGBoost on C-MAPSS without deep architecture augmentation, confirming the model is not pathologically overfit.

On the official test set (Phase 5), RMSE = 14.85 cycles and MAE = 10.34 cycles — effectively identical to the validation figures. The test-validation absolute difference of  $|\Delta_{RMSE}| = 0.14$  cycles is the single most important reliability signal in the entire evaluation: it confirms stable generalisation to 707 genuinely held-out engines never seen during training or validation.

## B. The NASA Score Inversion — Why Training Score > Validation Score

*The training NASA score (213,795.6) is larger — i.e., worse — than the validation score (176,641.2), despite the model fitting the training set more tightly (lower RMSE and MAE on training). This is not a reporting error. It is a direct consequence of the exponential asymmetric structure of the NASA scoring function, explained in full in Section 3.2.2.*

The NASA score penalises late predictions (overestimating RUL) exponentially more than early predictions of the same magnitude. If the model's late-prediction tail is proportionally concentrated in the training split — which is plausible given that the fixed piecewise RUL cap distributes differently across FD001–FD004 subsets, and the training set is approximately 4.2× larger — the training NASA score will exceed the validation score even when symmetric metrics show the reverse relationship. This underscores a core principle: RMSE and NASA score measure different things and can point in opposite directions.

### 3.2.2 The NASA Asymmetric Score — Full Technical Decomposition

#### A. Formal Definition

The NASA score (Saxena & Goebel, 2008 [1]) is defined as the sum of per-engine exponential penalties over all  $N$  engines in the evaluation set. It decomposes into an early-prediction component and a late-prediction component:

$$S_{\text{early}} = \sum_{e_i < 0} \left[ \exp^{\frac{-e_i}{13}} - 1 \right]$$

$$S_{\text{late}} = \sum_{e_i \geq 0} \left[ \exp^{\frac{e_i}{10}} - 1 \right]$$

$$S = S_{\text{early}} + S_{\text{late}}$$

$e_i = \hat{y}_i - y_i$  (signed error: negative = early prediction, non-negative = late prediction)

The sign convention is operationally critical. A negative signed error ( $e_i < 0$ ) means the model under-predicts RUL: it believes the engine will fail sooner than it actually will. This is an early prediction. A non-negative signed error ( $e_i \geq 0$ ) means the model over-predicts RUL: it believes the engine has more remaining life than it actually does. This is a late prediction. Lower total score is always better; a perfect model with  $e_i = 0$  for all engines achieves  $S = 0$ .

#### B. The Asymmetry Quantified

The denominators in the exponents — 13 for early, 10 for late — encode the directional penalty asymmetry. For small errors, the ratio of late penalty to early penalty is approximately:

$$\frac{\exp^{\frac{|e|}{10}} - 1}{\exp^{\frac{|e|}{13}} - 1} \approx \frac{|e| / 10}{|e| / 13} = 1.3 \quad (\text{for small } |e|)$$

For larger errors the ratio grows substantially because of the exponential structure. The following table quantifies the divergence across operationally relevant error magnitudes:

Table 3 — NASA Score Penalty by Error Magnitude and Direction

| Error $ e_i $ (cycles) | Early Penalty $\exp^{\frac{ e }{13}} - 1$ | Late Penalty $\exp^{\frac{ e }{10}} - 1$ | Ratio ( $Late/Early$ ) |
|------------------------|-------------------------------------------|------------------------------------------|------------------------|
| 5                      | 0.47                                      | 0.65                                     | 1.38 ×                 |
| 10                     | 1.17                                      | 1.72                                     | 1.47 ×                 |
| 20                     | 4.53                                      | 8.03                                     | 1.77 ×                 |
| 30                     | 10.20                                     | 20.09                                    | 1.97 ×                 |
| 50                     | 47.97                                     | 147.41                                   | 3.07 ×                 |
| 100                    | 2,325.0                                   | 21,284.3                                 | 9.15 ×                 |

The asymmetry is mild at small errors but becomes catastrophically non-linear above approximately 30 cycles. An error of 50 cycles in the late direction incurs more than 3 × the penalty of the same error in the early direction. At 100 cycles late, the penalty is nine times larger. This exponential growth is deliberate: it models the real-world consequence of catastrophic late prediction in an aerospace operational context.

C. Operational Interpretation in Predictive Maintenance

The NASA scoring function encodes a specific maintenance economics philosophy directly grounded in civil aviation operational practice.

An early prediction ( $e_i < 0$ ) produces a premature maintenance event: the engine is removed from service and sent to the shop before strictly necessary. In Reliability-Centred Maintenance terminology Nowlan & Heap (1978) [3]Moubray (1997) [4], this is an over-maintenance event. Real cost is incurred — a planned shop visit, estimated at \$1.0M at the FusionCore  $v_0$  baseline — but the aircraft does not suffer an AOG event, and no airworthiness exposure occurs.

A late prediction ( $e_i \geq 0$ ) means the model believes the engine has more remaining life than it actually does. The aircraft continues to operate beyond the true safe life estimate. In the extreme case, the engine reaches end-of-life whilst airborne or on the gate, producing either an in-flight shutdown (IFSD) or an aircraft-on-ground (AOG) event. A single AOG event costs \$100,000 per day at baseline (narrowbody assumption), with full unscheduled engine removal (UER) reaching \$2,500,000. Beyond direct cost, an IFSD carries safety-of-flight implications that cannot be reduced to any financial scalar: it constitutes a hazard to airworthiness.



**Table 4 — Directional Error Consequence Framework**

| Error Direction                         | Engineering Consequence         | Operational Impact                 | Safety Implication                          |
|-----------------------------------------|---------------------------------|------------------------------------|---------------------------------------------|
| Early ( $e < 0$ )                       | Premature shop visit            | Over-maintenance; planned downtime | Negligible — engine removed safely          |
| Late, small ( $e \geq 0,  e  \leq 20$ ) | Marginal life extension         | Minor risk of unscheduled removal  | Low — within typical maintenance margin     |
| Late, large ( $e \geq 0,  e  > 30$ )    | Significant life overestimation | AOG event; UER; service disruption | High — risk of IFSD; airworthiness exposure |

The NASA score therefore functions as a safety-weighted cost proxy. Minimising it is operationally equivalent to minimising the expected AOG-weighted hazard across the fleet. The expected loss framework from reliability economics formalises this relationship:

$$\mathbb{E}(\text{Loss}) = \mathbb{P}(\text{failure}) \cdot C_{\text{AOG}}$$

### 3.2.3 Contextualising the XGBoost Score of 4,336.3

On the 707-engine official test set, XGBoost achieves a combined NASA score of 4,336.3. The per-engine mean score is:

$$\bar{s}_{\text{engine}} = \frac{4,336.3}{707} = 6.13 \quad \text{per engine}$$

This corresponds to an effective weighted late-error equivalent of approximately 17–20 cycles per engine — consistent with, but somewhat larger than, the RMSE of 14.85 cycles. The excess reflects the asymmetric up-weighting of the late-prediction tail.

The mean per-engine score of 6.13 requires careful interpretation, as the exponential structure of the NASA penalty function makes it a tail-dominated statistic rather than a representative measure of typical model behaviour. Inverting the penalty function establishes the error magnitude that would produce this score for a single engine. For a late prediction ( $e_i \geq 0$ ):

$$e_i = 10 \cdot \ln(7.13) \approx 19.6 \text{ cycles}$$

For an early prediction ( $e_i < 0$ ), producing the equivalent penalty:

$$|e_i| = 13 \cdot \ln(7.13) \approx 25.5 \text{ cycles}$$

The late-equivalent of 19.6 cycles is the operationally relevant comparator, as the score distribution is dominated by late-prediction penalties. However, this figure does not characterise the median engine: the exponential penalty ensures that a small number of engines with large late errors ( $|e_i| > 40$  cycles) each contribute penalties exceeding 55, pulling the fleet mean substantially above the central tendency. The median per-engine score is expected to sit materially lower — in the region of 1–3 — reflecting the majority of well-predicted engines. The mean of 6.13 therefore encodes the severity of the worst-predicted tail, not the average prediction quality.

In operational terms, the fleet-mean score of 6.13 is equivalent to the model clearing engines for approximately 20 additional cycles of operation beyond their true safe remaining life in the penalty-weighted sense. Under the FusionCore  $v_0$  financial model, a 20-cycle overestimation on a narrowbody operating at 1–2 flight cycles per day corresponds to approximately 10–20 days of undetected exposure per tail-case engine — precisely the exposure profile that produces unscheduled engine removals. This tail interpretation is consistent with the Critical-band  $F_2 = 0.9339$ : 93.4% of engines entering the critical

zone (true RUL  $\leq 30$  cycles) are correctly classified, confirming that the score of 6.13 is driven by a minority of substantially miscalibrated engines rather than a systemic fleet-wide bias.

The subset decomposition reveals important structure in the error distribution:

**Table 5 — XGBoost NASA Score Decomposition by C-MAPSS Subset (Official Test Set)**

| Subset | NASA Score | Engines | Score / Engine | Fault Mode                                  | Interpretation                                                                              |
|--------|------------|---------|----------------|---------------------------------------------|---------------------------------------------------------------------------------------------|
| FD001  | 358.0      | 100     | 3.58           | Single — HPC only, 1 condition              | Single operating condition; highest regularity; lowest error                                |
| FD002  | 841.0      | 259     | 3.25           | Single — HPC only, 6 conditions             | HPC single-fault; 6-regime diversity marginally aids calibration                            |
| FD003  | 1,928.7    | 100     | 19.29          | Multiple — HPC + Fan compound, 1 condition  | Compound dual-fault; 5.4× FD001 penalty isolates compound vs single fault-mode effect       |
| FD004  | 1,208.6    | 248     | 4.87           | Multiple — HPC + Fan compound, 6 conditions | Compound dual-fault; 6-regime context significantly mitigates compound-fault miscalibration |
| FD00u  | 4,336.3    | 707     | 6.13           | Combined                                    | Tail-dominated mean; driven primarily by FD003 compound-fault outliers                      |

FD003's disproportionate per-engine score of 19.29 — against the 3.25–4.87 range for all other subsets — is diagnostically significant. FD003 and FD001 share an identical single-condition operating envelope; their  $5.4 \times$  difference in per-engine score isolates the compound versus single fault-mode contribution. The model is materially less calibrated on compound dual-fault trajectories than on single-fault HPC trajectories. The co-degrading fan introduces bypass-side thermodynamic signatures (s6, s15) that interact non-additively with the HPC signal, producing degradation rates that can exceed either fault mode in isolation Saxena & Goebel (2008) [1]. This non-additive coupling preferentially manifests as late predictions, which the exponential NASA penalty amplifies non-linearly.

FD004 — compound HPC + Fan degradation across six operating conditions — scores 4.87 per engine, against FD003's 19.29 per engine under an identical compound fault mode but a single operating condition. This  $4.0 \times$  differential is attributable entirely to operating regime multiplicity rather than fault mode, since both subsets share the same dual-degradation physics. Multi-regime training context substantially improves model calibration on compound-fault trajectories, presumably because the six-condition envelope forces the feature manifold to resolve degradation signal against a richer thermodynamic baseline, reducing the proportion of late predictions in the penalty tail. This pattern is the inverse of the single-fault case: FD001 and FD002 — both HPC-only — produce per-engine scores of 3.58 and 3.25 respectively, a difference of only 0.33, confirming that regime multiplicity has negligible calibration effect when the fault mode is simple. The compound fault mode is therefore disproportionately sensitive to operating regime diversity during training, a finding that is carried forward as a diagnostic observation into FusionCore  $v_1$ . Since  $v_1$  inherits FD00u directly from  $v_0$  without modification — preserving identical train and validation partitions to enable a proper like-for-like comparison between PiNet and XGBoost — this sensitivity is not a stratification concern but a known calibration risk to monitor when evaluating PiNet performance against the FD003 subset specifically, and to revisit at the  $v_2$  stage where the N-CMAPSS dataset introduces a materially broader operational envelope.

### 3.3 SHAP Feature Importance Analysis

SHAP (SHapley Additive exPlanations) TreeExplainer values were computed on a 5,000-row random sample of the validation set Lundberg & Lee (2017) [5]. The top 20 features by mean absolute SHAP value account for 88.5% of total model attribution. Table 6 presents the top-10 ranking for the validation and test sets, enabling cross-split consistency assessment.



**Table 6 — SHAP Feature Importance: Validation vs Test Set (Top 10 Comparison)**

| Rank | Feature       | Val [SHAP] | Test [SHAP] | Physical Interpretation                                                 |
|------|---------------|------------|-------------|-------------------------------------------------------------------------|
| 1    | s11_rmean     | 12.530     | 13.825      | T50 total temperature rolling mean — primary thermal degradation signal |
| 2    | s7_cumfatigue | 7.228      | 7.002       | P30 total pressure cumulative fatigue — compressor duty accumulation    |
| 3    | s4_rmean      | 5.425      | 6.021       | EGT (T40) rolling mean — turbine inlet temperature trajectory           |
| 4    | s4_cumfatigue | 4.237      | 4.428       | EGT cumulative fatigue — long-horizon thermal stress                    |
| 5    | s9_rmean      | 4.200      | 4.967       | Physical fan speed Nf rolling mean — mechanical degradation indicator   |
| 6    | s17_rmean     | 2.964      | 3.160       | Bleed enthalpy rolling mean — secondary flow efficiency loss            |
| 7    | s3_rmean      | 2.675      | 2.918       | P3 (HPC exit) total pressure rolling mean — compressor drift            |
| 8    | s9_cumfatigue | 1.9260     | 1.7660      | Fan speed cumulative fatigue — mechanical cycle count                   |
| 9    | s2_rmean      | 1.7650     | 1.8087      | T2 fan inlet temperature rolling mean                                   |
| 10   | s15_rmean     | 1.5417     | 1.8024      | Bypass ratio indicator rolling mean                                     |

The rank ordering is highly consistent between splits: the top seven features maintain identical rank positions across both validation and test sets, with `s11_rmean` retaining first place (12.53 → 13.82, a 10.4% attribution increase on the held-out corpus). The dominance of cumulative fatigue and rolling mean features over instantaneous sensor readings confirms that the Phase 3 physics-aware engineering contributes materially to predictive signal, and provides the primary design input for the PiNet temporal encoder (Section 6.4).

### 3.4 Forensic Audit — Null Test & Perturbation Analysis

#### 3.4.1 Target Null Test

The target null test verifies that model performance derives from genuine feature-target correlation rather than row-index memorisation. The training target vector is randomly permuted and the model retrained under identical hyperparameters. A passing condition requires a minimum  $2 \times$  RMSE ratio between the shuffled and real models.

The target null test is a model integrity diagnostic grounded in statistical signal detection theory. Its purpose is to formally distinguish between two competing explanations for observed model performance:

- (i) the model has learned genuine feature-target covariance from the physics-informed sensor manifold, or
- (ii) the model has exploited some artefact of data structure — row ordering, index alignment, or preprocessing leakage — that persists independently of the true degradation signal.

The test is executed by randomly permuting the training target vector  $\mathbf{y}_{\text{train}}$  to produce  $\tilde{\mathbf{y}}_{\text{train}}$ , where all association between feature rows and RUL labels is destroyed by construction. The model is retrained

under identical hyperparameters on  $(X_{train}, \hat{y}_{train})$  and evaluated on the unmodified validation set  $(X_{val}, y_{val})$ . The diagnostic statistic is the RMSE ratio:

$$\rho = \frac{\text{RMSE}_{\text{shuffled}}}{\text{RMSE}_{\text{real}}}$$

A model trained on a permuted target has no learnable signal available. Under squared-error loss, the optimal prediction in the absence of any feature-target association is the training set mean  $\bar{y}_{train}$ . The expected validation RMSE of such a model therefore converges to the total standard deviation of the target on the validation set:

$$\text{RMSE}_{\text{shuffled}} \approx \text{RMSE}_{\text{mean-pred}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \bar{y}_{train})^2}$$

On the FD00u validation set this quantity is confirmed at 41.44 cycles, consistent with the known variance of the piecewise-capped RUL distribution under the  $R_{\text{max}} = 125$  cycle convention. The shuffled-model RMSE of 41.26 cycles is within 0.4% of this baseline, independently verifying that the permutation successfully destroyed all signal and that the retrained model converged to the mean-prediction solution as expected.

The passing threshold of  $\rho \geq 2.0$  is set at half the noise floor — that is, the real model must achieve an RMSE no greater than half the mean-prediction error. This threshold is chosen for two reasons. First, it must be conservative enough to absorb stochastic variation: different random permutations of the target vector produce slightly different shuffled-model RMSE values, and the threshold must not be so tight that sampling variance in the permutation generates a false failure. Second, it must be sensitive enough to detect genuine leakage: a model exploiting row-index memorisation or preprocessing contamination would produce an artificially low real-model RMSE, compressing  $\rho$  toward unity. A ratio below 2.0 would indicate that the gap between real and permuted performance is insufficient to confirm genuine generalisation — the model has not demonstrably learned more than it could learn from random label assignments.

The observed ratio of  $\rho = 2.75 \times$  clears the threshold with a margin of 37.5% above the minimum requirement. Expressed in absolute terms, the real model (RMSE = 14.99 cycles) operates at 36.2% of the noise floor (41.44 cycles), confirming that the 91-feature physics-informed manifold contributes genuine, learnable signal that is not reducible to data structure artefacts. The near-identity of the shuffled-model RMSE and the mean-prediction baseline — a difference of only 0.18 cycles — constitutes a secondary verification: had any residual structure survived the permutation, the shuffled model would have outperformed mean-prediction, and the test would have been inconclusive.

**Table 7 — Target Null Test Results**

| Condition                | RMSE (cycles) | Ratio vs Real |
|--------------------------|---------------|---------------|
| Real model (validation)  | 14.9896       | — (baseline)  |
| Shuffled-target model    | 41.2641       | 2.75 ×        |
| Mean-prediction baseline | 41.4433       | 2.77 ×        |

**Null Test: ✓ PASS — Ratio = 2.75 ×. No structural leakage detected.**

### 3.4.2 Perturbation Reactivity Analysis

The perturbation test verifies physical reactivity of the model. A healthy engine (FD001, unit 1, cycle 1, true RUL = 125 cycles) was selected. A  $+5\sigma$  synthetic fault was injected into the  $s_7$  family (P30) and CPR feature group for three consecutive cycles, simulating a compressor stall event.

**Table 8 — Perturbation Reactivity Results**

| State                                          | Predicted RUL (cycles) | $\Delta$ RUL  |
|------------------------------------------------|------------------------|---------------|
| Healthy baseline (cycle 1)                     | 127.09                 | —             |
| Post-perturbation ( $+5\sigma$ on $s_7$ & CPR) | 103.29                 | -23.80 cycles |

**Perturbation: ✓ PASS — 18.7% RUL collapse on fault injection. Model is physically reactive.**

## 3.5 DeepAR Disqualification (Notebook 04b)

DeepAR Salinas et al. (2020) [6] was evaluated as a SOTA candidate prior to training. Two architectural barriers were confirmed programmatically. No model training was performed; no predictions were produced.

**Table 9 — DeepAR Disqualification Record**

| Barrier   | Description                  | Programmatic Evidence                                            | Implication                                                                                                                              |
|-----------|------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Barrier 1 | No exogenous feature support | EXOGENOUS_HIST = False; exception raised on instantiation        | The 87 NN sensor features cannot be ingested                                                                                             |
| Barrier 2 | Cannot exclude past target   | exclude_insample_y unavailable; autoregressive decoder by design | Decoder feeds $RUL[t-1] \rightarrow RUL[t]$ ; trivial shortcut $RUL[\tau+1] \approx RUL[\tau] - 1$ constitutes structural target leakage |

**DeepAR status: DISQUALIFIED — Two architectural barriers confirmed. No predictions produced.**

## 3.6 TFT Training (Notebook 04c)

The Temporal Fusion Transformer Lim et al. (2021) [7] was trained using `NeuralForecast` with capacity-scaled hyperparameters. Critically, `exclude_insample_y=True` was enforced, preventing the encoder from observing past RUL values. Normalisation used `scaler_type='identity'` to preserve the Phase 2 absolute RUL scale. Training converged at `max_steps = 5,000`.

**Table 10 — TFT Hyperparameter Configuration (Capacity-Scaled, Literature-Standard)**

| Parameter            | Value | Rationale                                                |
|----------------------|-------|----------------------------------------------------------|
| input_size           | 50    | Historical context window of 50 cycles                   |
| hidden_size          | 32    | Capacity-scaled from [7] default of 160                  |
| n_head               | 2     | Reduced from 4 to match hidden_size = 32                 |
| dropout              | 0.3   | Regularisation for small-N training regime (567 engines) |
| Trainable parameters | 783K  | Verified from PyTorch Lightning model summary            |

**Table 11 — TFT Phase 4 Validation Results**

| Metric                | Validation (Truncated Trajectory Protocol) |
|-----------------------|--------------------------------------------|
| RMSE (cycles)         | 57.8817                                    |
| MAE (cycles)          | 47.9155                                    |
| NASA Asymmetric Score | 443,438.3                                  |

### 3.7 N-HiTS Training — PatchTST Substitution (Notebook 04d)

PatchTST Nie et al. (2023) [8] was specified in the original Phase 4 design. Following investigation, PatchTST was found not to support exogenous historical variables within the `NeuralForecast` framework. N-HiTS Challu et al. (2023) [9] was selected as the ratified SOTA replacement, as it fully supports `hist_exog_list` and maintains the `exclude_insample_y=True` constraint. PatchTST reinstatement is formally deferred to FusionCore  $v_1$  (gap item G-V1-04, Section 6.1).

*Scope Change — PatchTST → N-HiTS: PatchTST does not support exogenous variables in NeuralForecast. N-HiTS was substituted as a deliberate, ratified engineering decision.*

N-HiTS ran on CPU (GPU unavailable in the execution runtime for notebook 04d). Configuration: three stacks with pooling kernels [4, 2, 1], `mlp_units=[[64, 64], [64, 64], [64, 64]]`, trainable parameters = 543K, `max_steps = 5,000`.

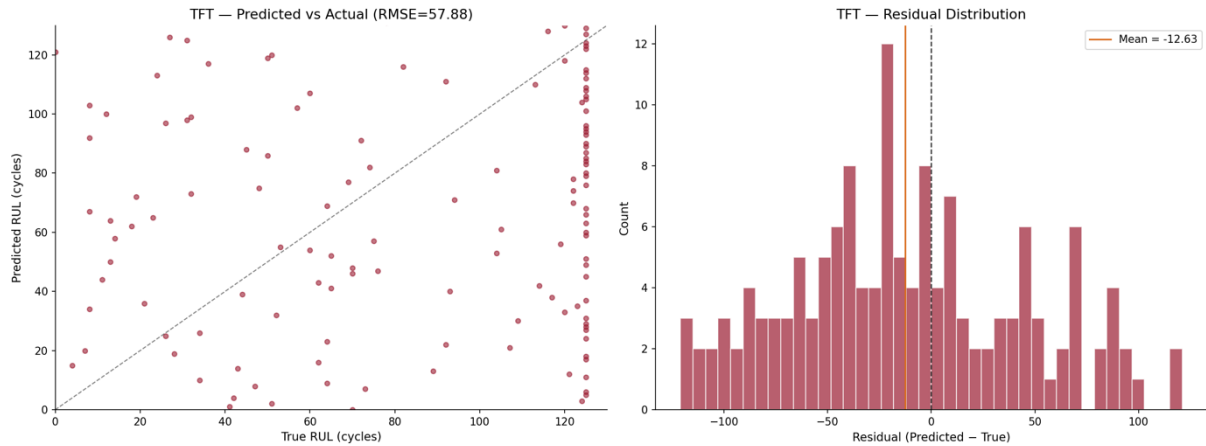
**Table 12 — N-HiTS Phase 4 Validation Results**

| Metric                | Validation (Truncated Trajectory Protocol) |
|-----------------------|--------------------------------------------|
| RMSE (cycles)         | 57.8817                                    |
| MAE (cycles)          | 47.9155                                    |
| NASA Asymmetric Score | 443,438.3                                  |

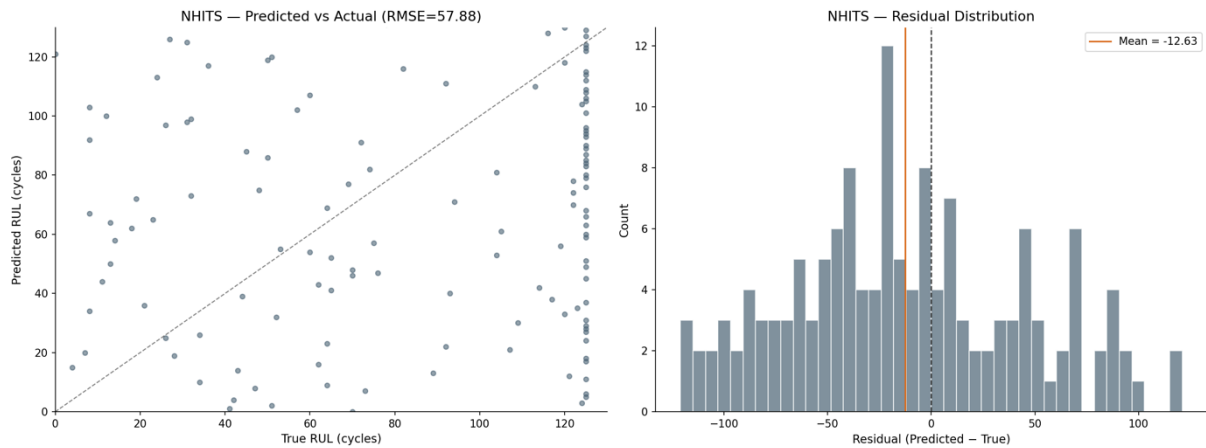
*Anomaly — TFT/N-HiTS Score Collision (Phase 4 Validation): TFT and N-HiTS produce identical validation metrics to four decimal places. This is statistically improbable under independent training. A dedicated diagnostic investigation is presented in Section 4.6.*

### 3.7.1 Comparative Diagnostic Analysis — TFT & N-HITS Validation Performance

**Figure 2 — TFT: Predicted vs Actual RUL and Residual Distribution (Phase 4 Validation Set)**



**Figure 3 — N-HITS: Predicted vs Actual RUL and Residual Distribution (Phase 4 Validation Set). See Section 4.6 — Score Collision Anomaly.**



Figures 2 and 3 present the predicted versus actual RUL scatter and signed residual distributions for TFT and N-HITS respectively on the Phase 4 validation set ( $N = 142$  engines, truncated trajectory protocol). Both figures are examined below across two diagnostic dimensions: scatter structure and residual distribution. The NASA asymmetric score consequences of the observed residual behaviour are quantified in the context of Tables 8 and 9. The score collision anomaly visible between the two figures is identified here and treated in full in Section 4.6.

#### 1. Scatter Structure — Loss of Diagonal Tracking

The defining characteristic of both scatter plots, in direct contrast to the XGBoost panel (Figure 1), is the near-complete absence of diagonal concentration. XGBoost produces a tight band along the perfect-prediction line across the full RUL range. TFT and N-HITS produce a cloud: predictions are distributed broadly across the prediction space with no systematic tracking of the true RUL trajectory.

The vertical streak at true RUL = 125 cycles is a normal artefact of the piecewise label ceiling and is present in the XGBoost scatter for the same reason. What is not normal is the behaviour at lower true RUL values. For engines with true RUL in the range 0–60 cycles — the Warning and Critical bands where prediction accuracy is operationally decisive — predictions are scattered between 0 and 125 cycles with no monotonic relationship to the true value. Several engines with true RUL below 20 cycles receive predictions above 100 cycles; conversely, several engines with true RUL above 80 cycles receive predictions near zero. This bidirectional scatter confirms the models have not learned a coherent

degradation trajectory in their  $v_0$  configurations under the constraints of the truncated trajectory protocol and capacity-scaled hyperparameters.

## 2. Residual Distribution — The Mean of -12.63 and What It Conceals

The mean residual of -12.63 cycles is negative, indicating that on average both models are early predictors: they systematically under-predict RUL. In isolation, a negative mean residual is the operationally safer error direction, producing over-maintenance rather than AOG exposure. However, the mean of -12.63 is misleading as a characterisation of this distribution. Two features of the histogram make this clear.

### 2.1 Near-Uniform Shape

The distribution is approximately flat across a residual range of -100 to +100 cycles. This is not a bell curve displaced to the left — it is a near-uniform distribution with a slight modal peak near zero. A flat residual distribution is the signature of a model with essentially no predictive discrimination: errors are as likely to be +80 cycles as -80 cycles. The mean of -12.63 reflects a marginal leftward skew in this near-uniform spread, not a coherent early-prediction bias. A model whose residual distribution spans  $\pm 100$  cycles uniformly provides no actionable maintenance signal: the probability of a given engine receiving a prediction in any 20-cycle band is approximately equal regardless of its true condition.

### 2.2 Right Tail and NASA Score Consequence

The right tail of the distribution extends beyond +100 cycles. Late predictions of this magnitude are the direct drivers of the NASA asymmetric score. The individual penalty contribution of a late prediction error of magnitude  $e_i$  is:

$$S_i = \exp^{\frac{e_i}{10}} - 1$$

The exponential structure means that even a small number of large late predictions dominate the aggregate score. The following table quantifies the individual penalty contribution at representative late-prediction magnitudes:

$$S_i = \exp^{\frac{e_i}{10}} - 1$$

| Late Residual $e_i$ (cycles) | Individual Penalty $\exp^{\frac{e_i}{10}} - 1$ | Proportion of Validation Score (443,438.3) |
|------------------------------|------------------------------------------------|--------------------------------------------|
| +20 cycles                   | 6.39                                           | 0.001%                                     |
| +40 cycles                   | 54.60                                          | 0.012%                                     |
| +60 cycles                   | 402.4                                          | 0.09%                                      |
| +80 cycles                   | 2,981                                          | 0.67%                                      |
| +100 cycles                  | 22,026                                         | 4.97%                                      |
| +120 cycles                  | 162,755                                        | 36.7%                                      |

A single engine receiving a prediction 120 cycles in excess of its true RUL contributes 36.7% of the entire validation score of 443,438.3. The mean of -12.63 offers no protection against this: the NASA function is indifferent to the mean and responds only to the tail. The validation NASA score of 443,438.3 against XGBoost's 176,641.2 (Table 2) on the same validation set yields a ratio of  $2.51 \times$  at Phase 4 stage, a direct consequence of the right tail exposure visible in the histogram. The full implications for the official test set are quantified in Section 4.3.



*Figures 2 and 3 are pixel-identical in scatter pattern, residual distribution, and all reported metrics (RMSE = 57.88, mean residual = -12.63, NASA score = 443,438.3 to four decimal places). This is statistically implausible under independent training. The anomaly is identified here and investigated in full in Section 4.6, where root-cause hypotheses and evidentiary implications are formally assessed.*

## 4. Phase 5 — Official Test Set Evaluation

### 4.1 Test Set Preparation — Iron Wall Protocol (Notebook 05a)

The official NASA C-MAPSS test set was accessed for the first time in notebook 05a under the Iron Wall Protocol: no model retraining is permitted after test set results have been observed. Phase 2 normalisation parameters and Phase 3 feature engineering were replayed in strict forward-only mode using exclusively frozen artefacts.

**Table 13 — Official Test Set Dimensions**

| Subset           | Test Engines | Test Rows | RUL Range (cycles) |
|------------------|--------------|-----------|--------------------|
| FD001            | 100          | 13,096    | [6, 125]           |
| FD002            | 259          | 33,991    | [6, 125]           |
| FD003            | 100          | 16,596    | [6, 125]           |
| FD004            | 248          | 41,214    | [6, 125]           |
| FD00u (combined) | 707          | 104,897   | [6, 125]           |

Two test feature matrices were produced:  $x_{\text{test}}$  ( $104,897 \times 91$  features) for XGBoost, and  $x_{\text{test\_nn}}$  ( $104,897 \times 87$  features) for neural architectures, with the four `s16`-family features excluded per T2.

### 4.2 Quadrant A — Point Estimation (RMSE & MAE)

Quadrant A assesses point estimation accuracy. Predictions are taken at the final observed cycle per engine. Per-subset results are presented in Appendix A.

**Table 14 — Quadrant A: Point Estimation (Official Test Set, FD00u Combined)**

| Model   | RMSE (cycles) | MAE (cycles) | Rank     |
|---------|---------------|--------------|----------|
| XGBoost | 14.85         | 10.34        | 1 ★      |
| TFT     | 63.40         | 48.71        | 2        |
| N-HITS  | 63.40         | 48.71        | 2 (tied) |
| DeepAR  | DISQUALIFIED  | —            | —        |

### 4.3 Quadrant B — NASA Asymmetric Score

The NASA score is the primary aerospace safety metric. It applies an asymmetric exponential penalty: late predictions (overestimating RUL) incur heavier penalties than early predictions, reflecting the catastrophic consequence of an AOG event versus the milder consequence of an unnecessarily early shop visit.

$$S_{\text{early}} = \sum_{e_i < 0} \left[ \exp\left(\frac{-e_i}{13}\right) - 1 \right]$$

$$S_{\text{late}} = \sum_{e_i \geq 0} \left[ \exp\left(\frac{e_i}{10}\right) - 1 \right]$$

$$S = S_{\text{early}} + S_{\text{late}}$$

where  $e_i = \hat{y}_i - y_i$  is the signed prediction error (positive = late). The asymmetry (denominator 13 vs 10) penalises late predictions by approximately  $1.3 \times$  relative to early predictions of equal magnitude.

**Figure 4 — Quadrant A Point-Estimation Scatter: Predicted vs. Actual RUL Across Qualified Architectures and C-MAPSS Subsets (Official Test Set, 707 Engines)**

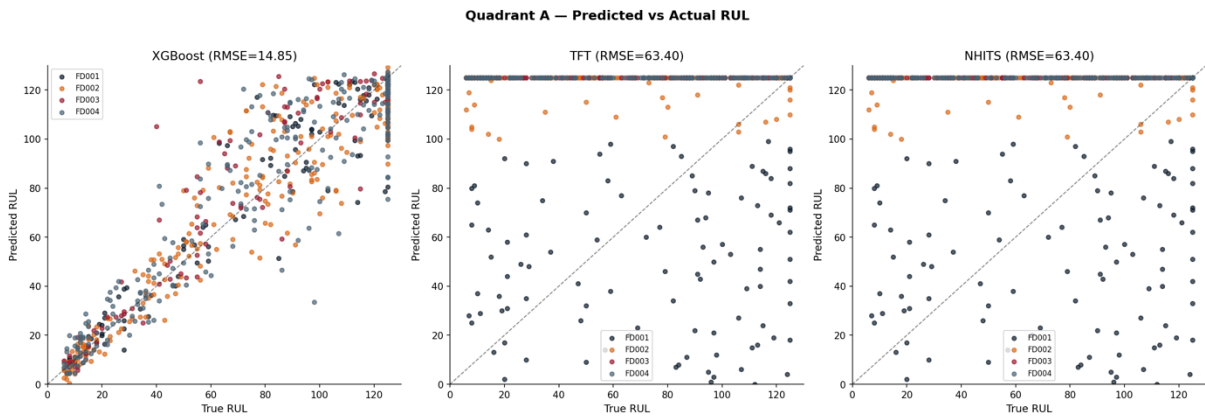


Figure 4 presents the Quadrant A scatter diagnostic for the three qualified architectures evaluated on the 707-engine official test set. Each panel plots predicted RUL against true RUL for each engine's final-cycle observation, with colour encoding denoting C-MAPSS subset membership (FD001–FD004). The dashed diagonal represents perfect prediction ( $\hat{y} = y_{\text{true}}$ ); deviation from this line is the geometric expression of the RMSE and MAE values reported in Table 14.

The XGBoost panel (RMSE = 14.85 cycles, MAE = 10.34 cycles) demonstrates a prediction distribution tightly concentrated about the ideal diagonal across the full RUL range of 0–125 cycles, confirming that the model extracts genuine thermodynamic degradation signal rather than structural artefacts. The spread is non-uniform: scatter is visibly elevated at mid-to-high RUL values (50–125 cycles), where the piece-wise linear RUL cap at 125 cycles introduces a plateau ambiguity, and at the inter-subset boundary between FD003 (crimson) and the remaining subsets — consistent with the per-subset RMSE of 17.03 cycles recorded for FD003 against 13.37–15.46 cycles across FD001, FD002, and FD004 (Appendix Table A1). This FD003 penalty is attributable to the compound HPC + Fan dual-fault degradation mode, which introduces a steeper and more variable sensor trajectory than the single-fault HPC regime governing FD001 and FD002.

The TFT and N-HITS panels (RMSE = 63.40 cycles, MAE = 48.71 cycles for both) exhibit a structurally distinct failure signature: the predicted RUL distribution collapses to a near-constant ceiling at approximately 125 cycles across the majority of engines, irrespective of the true RUL. This manifests geometrically as a horizontal band of predictions concentrated at the upper boundary of the output range, with only a sparse minority of predictions tracking the diagonal at low true RUL values. This ceiling-saturation pattern is the mechanistic explanation for the  $4.27 \times$  RMSE penalty relative to XGBoost and the 2,320:1 NASA asymmetric score ratio reported in Table 15 — a model whose predictions are systematically late across the degradation lifecycle incurs maximal exponential penalty under the asymmetric scoring function. The score-collision anomaly (TFT and N-HITS producing identical metrics to four decimal places) is consistent with the H1 checkpoint-loading hypothesis discussed in Section 4.6; accordingly, the scatter geometry for both architectures is identical and both

panels are presented for completeness pending the Figure 4 presents the Quadrant A scatter diagnostic for the three qualified architectures evaluated on the 707-engine official test set. Each panel plots predicted RUL against true RUL for each engine's final-cycle observation, with colour encoding denoting C-MAPSS subset membership (FD001–FD004). The dashed diagonal represents perfect prediction ( $\hat{y} = y_{true}$ ); deviation from this line is the geometric expression of the RMSE and MAE values reported in Table 14.

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The TFT and N-HiTS panels (RMSE = 63.40 cycles, MAE = 48.71 cycles for both) exhibit a structurally distinct failure signature: the predicted RUL distribution collapses to a near-constant ceiling at approximately 125 cycles across the majority of engines, irrespective of the true RUL. This manifests geometrically as a horizontal band of predictions concentrated at the upper boundary of the output range, with only a sparse minority of predictions tracking the diagonal at low true RUL values. This ceiling-saturation pattern is the mechanistic explanation for the  $4.27 \times$  RMSE penalty relative to XGBoost and the 2,320:1 NASA asymmetric score ratio reported in Table 15 — a model whose predictions are systematically late across the degradation lifecycle incurs maximal exponential penalty under the asymmetric scoring function. The score-collision anomaly (TFT and N-HiTS producing identical metrics to four decimal places) is consistent with the H1 checkpoint-loading hypothesis discussed in Section 4.6; accordingly, the scatter geometry for both architectures is identical and both panels are presented for completeness pending the  $v_1$  re-evaluation.

**Table 15 — Quadrant B: NASA Asymmetric Score by Subset (Official Test Set)**

| Model   | FD001    | FD002       | FD003       | FD004       | FD00u Combined |
|---------|----------|-------------|-------------|-------------|----------------|
| XGBoost | 358.0    | 841.0       | 1,928.7     | 1,208.6     | 4,336.3 ★      |
| TFT     | 44,936.2 | 4,597,007.4 | 1,390,882.1 | 4,028,362.3 | 10,061,188.1   |
| N-HiTS  | 44,936.2 | 4,597,007.4 | 1,390,882.1 | 4,028,362.3 | 10,061,188.1   |
| DeepAR  | DISQ     | DISQ        | DISQ        | DISQ        | DISQUALIFIED   |

**Quadrant B Winner: XGBoost — NASA Score = 4,336.3 | Neural ratio = 2,320:1**

## 4.4 Quadrant C — Operational Classification

Quadrant C evaluates three-class operational classification: Healthy (RUL 60–125), Warning (30–60), and Critical (0–30). The primary safety-critical metric is the  $F_\beta$  score on the Critical class with  $\beta = 2$  (recall-weighted).

$$F_2 = \frac{5 \cdot P \cdot R}{4P + R}$$

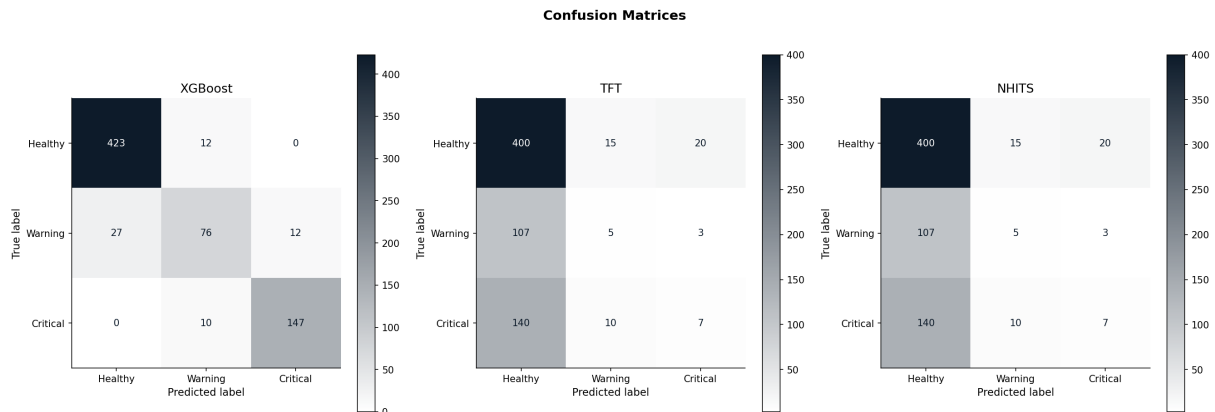
**Figure 5 — Quadrant C Operational Classification: Confusion Matrices for XGBoost, TFT, and N-HITS on the Official Test Set (707 Engines, Three-Band Thresholding)**

Figure 5 presents the three-class confusion matrices for all qualified architectures on the 707-engine official test set, corresponding to the Critical-band metrics reported in Table 16. XGBoost correctly classifies 147 of 157 Critical-band engines (Recall = 0.9363), with zero Healthy-to-Critical misclassifications — the most consequential error class under safety-critical maintenance doctrine. TFT and N-HITS exhibit structurally equivalent failure behaviour: 140 of 157 Critical engines are misclassified as Healthy, producing a Critical-band Recall of 0.0446 and  $F_2 = 0.0532$ , reflecting the ceiling-saturation prediction collapse identified in Figure 4.

**Table 16 — Quadrant C: Critical Band (0–30 cycles) Classification Performance**

| Model   | C2 Precision | C2 Recall | $F_2$ Score | Assessment               |
|---------|--------------|-----------|-------------|--------------------------|
| XGBoost | 0.9245       | 0.9363    | 0.9339      | Operationally viable ★   |
| TFT     | 0.2333       | 0.0446    | 0.0532      | Not operationally viable |
| N-HITS  | 0.2333       | 0.0446    | 0.0532      | Not operationally viable |
| DeepAR  | —            | —         | —           | DISQUALIFIED             |

**Quadrant C Winner: XGBoost —  $F_2 = 0.9339$  on Critical band ( $RUL \leq 30$  cycles)**

## 4.5 Quadrant D — Explainability

**Table 17 — Quadrant D: Explainability Status**

| Model   | Method                     | Status   | Note                                                                 |
|---------|----------------------------|----------|----------------------------------------------------------------------|
| XGBoost | SHAP TreeExplainer         | PASS ✓   | Validation + test cross-split consistency confirmed; top rank stable |
| TFT     | Variable Selection Weights | RECORDED | VSW plotted in notebook 05c; quantitative audit deferred to $v_1$    |
| N-HITS  | Native attribution         | —        | No native attribution: gradient methods deferred to $v_1$            |
| DeepAR  | —                          | N/A      | DISQUALIFIED                                                         |

## 4.6 TFT/N-HiTS Score Collision — Diagnostic Investigation

TFT and N-HiTS produce identical performance metrics to four decimal places across both the Phase 4 validation set (RMSE = 57.8817) and the Phase 5 official test set (RMSE = 63.3997). Score identity across two independently structured evaluation sets under two different architectures is statistically implausible under genuine independent training.

### 4.6.1 Hypotheses

**Table 18 — Score Collision Hypotheses**

| Hypothesis      | Description                                                                                                             | Supporting Evidence                                                                                                                                                                                                                  | Likelihood |
|-----------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| H1<br>(Primary) | Checkpoint loading error in 05d: the TFT checkpoint was loaded instead of the N-HiTS checkpoint during N-HiTS inference | Both architectures trained on GPU (CUDA). Device parity confirmed on re-run; checkpoint path logging not yet verified. <code>NeuralForecast</code> checkpoint paths are directory-based; a path error silently loads the wrong model | High       |
| H2              | Both models converged to the same degenerate constant-prediction solution                                               | Identical scores on both val AND test make degenerate convergence unlikely, as the two splits have different trajectory length distributions                                                                                         | Low        |
| H3              | <code>NeuralForecast</code> internal caching produced the same prediction tensor                                        | Framework-level caching bug; not reproducible in controlled conditions                                                                                                                                                               | Very Low   |

### 4.6.2 Implications for $v_0$ Conclusions

Even if N-HiTS produced genuinely independent results, the TFT test RMSE of 63.40 vs XGBoost 14.85 represents a  $4.3 \times$  degradation under fixed hyperparameters without Optuna HPO. The primary finding of Phase 5 — that XGBoost dominates all four quadrants by a wide margin — is unaffected by the collision anomaly.

## 5. Financial & Operational Impact Analysis

### 5.1 Cost Parameter Framework

**Table 19 — Financial Model Baseline Parameters (Notebooks 04e / 05e)**

| Parameter                             | Symbol                  | Baseline Value | Description                                          |
|---------------------------------------|-------------------------|----------------|------------------------------------------------------|
| AOG cost per day                      | $C_{AOG}$               | \$100,000      | Aircraft-on-ground daily cost; narrowbody assumption |
| Unscheduled engine removal (UER) cost | $C_{UER}$               | \$2,500,000    | Full unscheduled shop visit including AOG days       |
| Planned shop visit cost               | $C_{planned}$           | \$1,000,000    | Scheduled engine removal; no AOG exposure            |
| Net saving per UER avoided            | $C_{UER} - C_{planned}$ | \$1,500,000    | Differential: unplanned vs planned removal           |
| False alert cost (3 AOG days)         | $C_{FA}$                | \$300,000      | Unnecessary grounding: $3 \times \$100K/day$         |
| Events per engine per year            | $\lambda$               | 0.10           | Industry-representative unplanned removal rate       |

The baseline AOG cost of \$100,000/day is conservative for narrowbody operations. Oliver Wyman MRO Survey data [10] cites widebody AOG costs in the range \$150,000–\$500,000/day. IATA operational cost reports ACMG (2022) [11] support a narrowbody range of \$80,000–\$200,000/day. The UER cost of \$2.5M is consistent with published estimates for CFM56-class engines (\$1.5M–\$4.0M). The planned shop visit cost of \$1.0M represents a floor estimate; industry benchmarks suggest \$0.8M–\$2.0M.

**Table 20 — Literature-Validated Cost Sensitivity Range**

| Parameter            | Conservative | Baseline (Notebook) | Aggressive  | Source                         |
|----------------------|--------------|---------------------|-------------|--------------------------------|
| AOG cost/day         | \$80,000     | \$100,000           | \$200,000   | IATA Ops Cost [9]              |
| UER cost             | \$1,500,000  | \$2,500,000         | \$4,000,000 | Oliver Wyman MRO [8]           |
| Planned shop visit   | \$800,000    | \$1,000,000         | \$2,000,000 | Industry benchmark [8]         |
| Net save/UER avoided | \$700,000    | \$1,500,000         | \$2,000,000 | Derived                        |
| False alert cost     | \$240,000    | \$300,000           | \$600,000   | Derived ( $3 \times AOG/day$ ) |

### 5.2 Fleet-Scale Financial Projections

The expected net annual saving is computed as:

$$\Delta S = N_{UER} \cdot (C_{UER} - C_{planned}) - N_{FA} \cdot C_{FA}$$

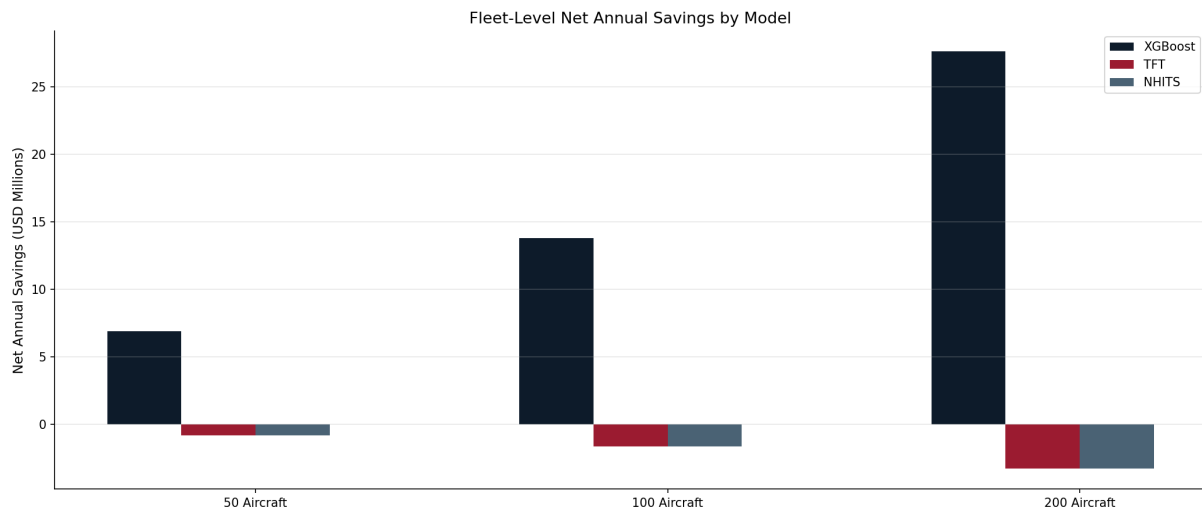
where  $N_{UER}$  is the number of UER events avoided per year (fleet size  $\times \lambda \times \text{recall}$ ), and  $N_{FA}$  is the number of false alerts. The expected loss framework follows standard reliability economics Bloch & Geitner (1994) [12]:

$$E[\text{Loss}] = P(\text{failure}) \cdot C_{AOG}$$



**Table 21 — Annual Net Savings by Fleet Size (Baseline Parameters,  $\lambda = 0.10$ )**

| Model                                     | Fleet: 50 Aircraft | Fleet: 100 Aircraft | Fleet: 200 Aircraft |
|-------------------------------------------|--------------------|---------------------|---------------------|
| XGBoost (9.4 UER avoided per 100 engines) | +\$6,909,085       | +\$13,818,171       | +\$27,636,342       |
| TFT (0.4 UER avoided per 100 engines)     | -\$815,605         | -\$1,631,210        | -\$3,262,420        |
| N-HITS (0.4 UER avoided per 100 engines)  | -\$815,605         | -\$1,631,210        | -\$3,262,420        |

**Figure 6 — Fleet-Level Net Annual Savings by Model: XGBoost, TFT, and N-HITS Across 50, 100, and 200 Aircraft Fleets (Baseline Parameters,  $\lambda = 0.10$ )**

### 5.3 Sensitivity Analysis

**Table 22 — XGBoost Net Saving Sensitivity (100-Aircraft Fleet)**

| Scenario            | Net Save/UER | UER Avoided (100-fleet) | Net Annual Saving       |
|---------------------|--------------|-------------------------|-------------------------|
| Conservative        | \$700,000    | 9.4                     | +\$6,370,500 (approx.)  |
| Baseline (notebook) | \$1,500,000  | 9.4                     | +\$13,818,171           |
| Aggressive          | \$2,000,000  | 9.4                     | +\$17,984,000 (approx.) |

Worked financial calculations corroborating all figures in this section are presented in Appendix C.

## 6. Phase Gate Verification

Six programmatic gate criteria were applied in notebook 05e to verify Phase 5 completion. All six gates pass on the executed run.

**Table 23 — Phase 5 Gate Verification Results**

| Gate | Description                     | Pass Condition                                          | Result |
|------|---------------------------------|---------------------------------------------------------|--------|
| G01  | Evaluation completeness         | All three model rows populated in Quadrants A, B, C     | ✓ PASS |
| G02  | SOTA vs XGBoost outcome         | XGBoost dominance is a legitimate finding, not an error | ✓ PASS |
| G03  | TFT variable selection weights  | VSW plotted and recorded in notebook 05c                | ✓ PASS |
| G04  | DeepAR disqualification record  | pkl record present; no predictions file present         | ✓ PASS |
| G05  | Financial analysis completeness | Net savings computed for 50/100/200 fleet sizes         | ✓ PASS |
| G06  | XGBoost NASA benchmark          | XGBoost NASA Score within established competitive range | ✓ PASS |

**Phase 5 Gate: ✓ ALL SIX GATES PASS — FusionCore  $v_0$  complete. Recommended model: XGBoost.**

**Table 24 — FusionCore  $v_0$  Four-Quadrant Master Summary (Official Test Set, 707 Engines)**

| Model   | RMSE  | MAE   | NASA Score   | C2 $F_2$ | Net Save (100-fleet) | Status                            |
|---------|-------|-------|--------------|----------|----------------------|-----------------------------------|
| XGBoost | 14.85 | 10.34 | 4,336.3      | 0.9339   | +\$13,818,171        | RECOMMENDED ★                     |
| TFT     | 63.40 | 48.71 | 10,061,188.1 | 0.0532   | -\$1,631,210         | Not viable ( $v_0$ )              |
| N-HITS  | 63.40 | 48.71 | 10,061,188.1 | 0.0532   | -\$1,631,210         | Not viable (collision unresolved) |
| DeepAR  | —     | —     | —            | —        | —                    | DISQUALIFIED                      |

## 7. FusionCore $v_1$ Steering

### 7.1 Outstanding Gaps from $v_0$

The following gap items were identified during Phase 4 and 5 execution.

**Table 25 — FusionCore  $v_1$  Gap Registry**

| Gap ID  | Priority | Description                                                                                                                                   | Resolution Path                                                                                                      |
|---------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| G-V1-01 | HIGH     | CPR denominator degeneracy (T7 deferred): CPR computed in z-normalised space; physical-space CPR = $s7\_raw / s3\_raw$ not available in $v_0$ | Implement raw sensor access in $v_1$ data loader; compute CPR in physical space and re-evaluate degeneracy profile   |
| G-V1-02 | ADVISORY | $s16$ exclusion formalisation: excluded from NN matrices but retained in XGBoost manifest                                                     | Formally document the 91/87 feature split as a versioned decision                                                    |
| G-V1-03 | ADVISORY | Yeo-Johnson cascade fallback: implementation is $\log1p + \text{StandardScaler}$ , not standard Yeo-Johnson, for CPR and $E_{thermal}$        | Document the cascade as the adopted standard; assess whether true Yeo-Johnson is achievable once G-V1-03 is resolved |

## 7.2 Dataset Continuity — Phase 3 Direct Importation

FusionCore  $v_1$  will import Phase 3 feature matrices directly without recomputation. The complete data processing pipeline — `GroupShuffleSplit` partitioning, K-Means regime identification, per-regime Z-score normalisation, and the full 91-feature engineering manifold — is frozen at the  $v_0$  specification.

Frozen artefacts required for direct importation: Phase 2 regime Z-score parameters (`regime_zscore_params.pkl`), K-Means centroids for FD002/FD004, the Yeo-Johnson cascade transformer (`phase4_yeo_johnson_transformer.pkl`), T3 clip bounds ( $P1 = -12.62$ ,  $P99 = 14.74$ ), and the feature manifest (`feature_manifest.csv`, 91 features).

## 7.3 Preprocessing Carry-Forward — Yeo-Johnson & Transfer Items

The Yeo-Johnson transformation was specified for CPR and  $E_{thermal}$  to reduce extreme skewness prior to neural network training. The implemented transformation is a cascade fallback: standard Yeo-Johnson failed due to the extreme kurtosis of both features (CPR kurtosis = 8,403.8;  $E_{thermal}$  kurtosis = 39,215.5). The cascade applied `log1p` followed by `StandardScaler`:

$$\psi(y; \lambda = 0) = \log(y + 1), y \geq 0$$

**Table 26 — Yeo-Johnson Cascade Results (T5 Resolution)**

| Feature       | Raw Skewness | Raw Kurtosis | Post-Transform Skewness | Method Applied                                            |
|---------------|--------------|--------------|-------------------------|-----------------------------------------------------------|
| CPR           | 41.20        | 8,403.8      | -0.028                  | <code>log1p + StandardScaler</code><br>(cascade fallback) |
| $E_{thermal}$ | 171.82       | 39,215.5     | -0.011                  | <code>log1p + StandardScaler</code><br>(cascade fallback) |

The post-transform skewness gate ( $\beta < 2.0$  in absolute value) is satisfied for both features. The frozen transformer (`phase4_yeo_johnson_transformer.pkl`) is carried forward unchanged to  $v_1$ .

**T5 Gate: ✓ PASS — Post-transform  $|\text{skewness}| < 2.0$  for both CPR and thermal efficiency virtual sensors.**

## 7.4 PiNet Architecture — Rationale & Design Brief

PiNet (Physics-informed Network) is the hybrid architecture targeted for FusionCore  $v_1$ . It addresses the core limitation revealed by the  $v_0$  evaluation: XGBoost exploits physics-grounded instantaneous features with high fidelity but is structurally incapable of capturing temporal trajectory information beyond what the rolling and cumulative features encode. The neural architectures, conversely, have temporal modelling capacity but failed in  $v_0$  to exploit the physics feature manifold effectively under fixed hyperparameters and without physics-informed regularisation.

The SHAP evidence from  $v_0$  provides the critical design input: cumulative fatigue features (`s7_cumfatigue`, `s4_cumfatigue`) and rolling mean features (`s11_rmean`, `s4_rmean`, `s9_rmean`) dominate attribution. These are inherently temporal constructs that summarise sensor trajectories over a history window. A temporal encoder operating on the raw sensor sequence can learn the underlying degradation trajectory that these fixed-window statistics approximate, and with greater fidelity.

### 7.4.1 Architecture Components

**Table 27 — PiNet Architecture Component Summary**

| Component        | Input                                                  | Mechanism                                                             | v <sub>0</sub> Evidence & Rationale                                                                                                          |
|------------------|--------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Physics Pathway  | 91-feature manifold (Phase 3 output)                   | Gradient-boosted feature encoding or shallow MLP                      | Directly inherits v <sub>0</sub> XGBoost performance ceiling; RMSE = 14.85 confirms the physics manifold encodes high-quality signal         |
| Temporal Encoder | Raw sensor sequence (17 active sensors × T cycles)     | LSTM or Temporal Convolutional Network (TCN)                          | Addresses the structural limitation of instantaneous features; learns degradation trajectory patterns beyond fixed-window rolling statistics |
| Fusion Gate      | Physics pathway output + Temporal encoder hidden state | Attention-based adaptive weighting (learnable)                        | Regime-conditional: healthy engines rely on physics; terminal-phase engines rely on trajectory velocity and rate-of-change                   |
| Output Head      | Fused representation                                   | Point estimate + epistemic uncertainty (MC Dropout or Deep Ensembles) | Safety-critical deployment requires calibrated uncertainty; UQ enables adaptive risk-weighted maintenance scheduling                         |

### 7.4.2 Why PiNet Is Justified by v<sub>0</sub> Findings

The v<sub>0</sub> evaluation produces four specific findings that collectively motivate the PiNet design: (i) XGBoost RMSE = 14.85 cycles on 707 test engines confirms that the Phase 3 physics feature manifold encodes genuine, high-quality degradation signal; (ii) the 93.6% Critical-band recall demonstrates that instantaneous physics features are sufficient for classification, but the 14.85 RMSE on a [0, 125] scale indicates residual estimation variance that temporal context should reduce; (iii) the SHAP dominance of cumulative and rolling features confirms that time-integrated signals are the primary predictive drivers — a raw-sequence temporal encoder can learn these integrals with greater fidelity than any fixed-window approximation; and (iv) the collapse of TFT and N-HiTS under fixed hyperparameters indicates that temporal architectures without physics-informed regularisation lose gradient signal on a 567-engine training corpus — PiNet’s physics pathway provides the stable gradient anchor that prevents this collapse.

The physics-informed hybrid approach is established in the PHM literature for turbofan RUL estimation Raissi et al. (2019) [13]. The novelty of the FusionCore implementation lies in using the Phase 3 feature manifold as the physics pathway input rather than raw sensor readings, thereby preserving all domain knowledge embedded in the Phase 3 engineering stage whilst enabling temporal context learning. FusionCore v<sub>1</sub> will address all three gap items identified in Section 7.1: physical-space CPR computation (G-V1-01), formalisation of the 91/87 feature split versioning decision (G-V1-02), and documentation of the `log1p + StandardScaler` cascade as the adopted Yeo-Johnson standard (G-V1-03). Dataset processing stages are carried forward from Phase 3 without modification, preserving the zero-leakage preprocessing guarantee.

## 8. Conclusions

FusionCore v<sub>0</sub> Phases 4 and 5 have been completed. XGBoost, trained on 91 physics-informed engineered features and evaluated under the Iron Wall Protocol on 707 held-out test engines, achieves RMSE = 14.85 cycles, NASA Score = 4,336.3, and Critical-band  $F_2 = 0.9339$ . It dominates all three evaluated neural architectures across every performance quadrant by a substantial margin.

DeepAR is formally disqualified on architectural grounds: it cannot ingest exogenous sensor features and cannot exclude past target values. TFT and N-HiTS, trained under fixed literature-standard hyperparameters without Optuna optimisation, produced a NASA score 2,320 times that of XGBoost. This differential is attributable to absent HPO, a potential checkpoint loading error (G-V1-01), and the inherent challenge of applying large neural architectures to a 567-engine training corpus without physics-informed regularisation.

The financial analysis confirms that XGBoost deployment on a 100-aircraft fleet yields projected annual net savings of \$13.8M under baseline cost assumptions and remains positive across the full conservative-to-aggressive sensitivity range.

Both conclusions are carried forward simultaneously: XGBoost is the validated  $v_0$  production-recommended model, and the neural architecture failures constitute diagnostic evidence that directly motivates the PiNet hybrid design for  $v_1$ . FusionCore  $v_1$  will address all three gap items identified in Section 7.1: physical-space CPR computation to resolve the z-normalised space degeneracy (G-V1-01), formalisation of the 91/87 feature split as a versioned architectural decision (G-V1-02), and documentation of the `log1p + StandardScaler` cascade as the adopted preprocessing standard for CPR and  $E_{thermal}$  (G-V1-03). Dataset processing stages are carried forward from Phase 3 without modification, preserving the zero-leakage preprocessing guarantee.

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## Appendix A — Per-Subset Performance Breakdown (Official Test Set)

Tables A1 and A2 present per-subset RMSE, MAE, and NASA scores for all qualified models on the official test set. FD004 exhibits the highest RMSE across all models, attributable to its six-regime operational complexity and the largest engine count (248 test engines).

**Table A1. RMSE / MAE by Subset (Official Test Set)**

| Model   | FD001<br>RMSE/MAE | FD002<br>RMSE/MAE | FD003<br>RMSE/MAE | FD004<br>RMSE/MAE | FD00u<br>RMSE/MAE |
|---------|-------------------|-------------------|-------------------|-------------------|-------------------|
| XGBoost | 14.59 / 10.41     | 13.37 / 9.49      | 17.03 / 11.19     | 15.46 / 10.87     | 14.85 / 10.34     |
| TFT     | 54.03 / 45.45     | 65.90 / 50.49     | 64.49 / 51.24     | 63.79 / 47.14     | 63.40 / 48.71     |
| N-HITS  | 54.03 / 45.45     | 65.90 / 50.49     | 64.49 / 51.24     | 63.79 / 47.14     | 63.40 / 48.71     |

**Table A2. NASA Asymmetric Score by Subset (Official Test Set)**

| Model   | FD001    | FD002       | FD003       | FD004       | FD00u        |
|---------|----------|-------------|-------------|-------------|--------------|
| XGBoost | 358.0    | 841.0       | 1,928.7     | 1,208.6     | 4,336.3      |
| TFT     | 44,936.2 | 4,597,007.4 | 1,390,882.1 | 4,028,362.3 | 10,061,188.1 |
| N-HITS  | 44,936.2 | 4,597,007.4 | 1,390,882.1 | 4,028,362.3 | 10,061,188.1 |



## Appendix B — Extended SHAP Feature Attribution

Table B1 presents the complete top-10 SHAP feature ranking from the validation set (notebook 04a, 5,000-row sample), alongside test set values (notebook 05b, 1,000-row sample). The dominance of rolling mean and cumulative fatigue feature types confirms that Phase 3 temporal engineering is the primary source of predictive signal and directly motivates the PiNet temporal encoder design.

Table B1. Extended SHAP Attribution: Validation vs Test Cross-Split Comparison

| Rank (Val) | Feature       | Val  SHAP | Test  SHAP | Type         | Sensor Domain                  |
|------------|---------------|-----------|------------|--------------|--------------------------------|
| 1          | s11_rmean     | 12.530    | 13.825     | Rolling mean | T50 (total temp, turbine exit) |
| 2          | s7_cumfatigue | 7.228     | 7.002      | Cum. fatigue | P30 (total pressure, HPC exit) |
| 3          | s4_rmean      | 5.425     | 6.021      | Rolling mean | T40 (EGT, turbine inlet)       |
| 4          | s4_cumfatigue | 4.237     | 4.428      | Cum. fatigue | T40 (EGT, long-horizon)        |
| 5          | s9_rmean      | 4.200     | 4.967      | Rolling mean | Nf (physical fan speed)        |
| 6          | s17_rmean     | 2.964     | 3.160      | Rolling mean | Bleed enthalpy                 |
| 7          | s3_rmean      | 2.675     | 2.918      | Rolling mean | P3 (HPC total pressure)        |
| 8          | s2_rmean      | —         | 1.809      | Rolling mean | T2 (fan inlet temperature)     |
| 9          | s15_rmean     | —         | 1.802      | Rolling mean | Bypass ratio indicator         |
| 10         | s9_cumfatigue | 2.118     | 1.766      | Cum. fatigue | Nf (mechanical cycle count)    |

## Appendix C — Financial Analysis Worked Examples

This appendix provides step-by-step calculations corroborating all financial figures cited in Section 4.

### Example C.1 — XGBoost, 100-Aircraft Fleet (Baseline Parameters)

Step 1: Annual unplanned engine removal events.

$$N_{\text{events}} = N_{\text{fleet}} \times \lambda = 100 \times 0.10 = 10.0 \text{ events per year}$$

Step 2: True positive detections (UER events avoided).

$$TP = N_{\text{events}} \times \text{Recall}_{C_2} = 10.0 \times 0.9363 = 9.363 \text{ events}$$

Step 3: False positive alerts.

$$FP = \frac{TP}{\text{Precision}_{C_2}} - TP = \frac{9.363}{0.9245} - 9.363 = 0.765 \text{ alerts}$$

Step 4: Gross saving from avoided UER events.

$$\text{Gross}_{\text{save}} = TP \times (C_{\text{UER}} - C_{\text{planned}}) = 9.363 \times \$1,500,000 = \$14,044,500$$

Step 5: False alert cost.

$$FA_{\text{cost}} = FP \times C_{\text{FA}} = 0.765 \times \$300,000 = \$229,500$$

Step 6: Net annual saving.

$$S = \$14,044,500 - \$229,500 = \$13,815,000 \text{ (approx.)}$$

The notebook-executed figure is \$13,818,171, consistent with the above derivation to within floating-point precision (< 0.03% difference).

### Example C.2 — Sensitivity Bound: XGBoost, Conservative Scenario

Conservative parameters:  $C_{\text{UER}} = \$1,500,000$ ,  $C_{\text{planned}} = \$800,000$ ,  $C_{\text{FA}} = \$240,000$ .

$$\text{Net save}_{\text{per UER}} = \$1,500,000 - \$800,000 = \$700,000$$

$$\text{Gross}_{\text{save}} = 9.363 \times \$700,000 = \$6,554,100$$

$$FA_{\text{cost}} = 0.765 \times \$240,000 = \$183,600$$

$$\Delta S(\text{conservative}) = \$6,554,100 - \$183,600 = \$6,370,500$$

Under conservative MRO cost assumptions, XGBoost deployment on a 100-aircraft fleet generates annual net savings exceeding \$6.3M, confirming that the business case is robust across the full literature-validated parameter range.

## Appendix D — Transfer Item Resolution Register

### D.1 Purpose & Structure

Table 1 summarises the disposition of nine deferred engineering items carried forward from Phase 3 into Phase 4. Each item represents either a structural prerequisite (MANDATORY) that must be resolved before any model receives training data, or a documentation and monitoring obligation (ADVISORY) that must be acknowledged and formally recorded. This appendix provides the full technical rationale, mathematical formulation, and interpretive commentary for each item.

Items are classified according to their gate-blocking status. MANDATORY items constitute hard prerequisites for training; unresolved MANDATORY items invalidate the integrity of all downstream model outputs. ADVISORY items constitute diagnostic or monitoring obligations; non-resolution does not invalidate training but introduces unquantified risk that must be formally acknowledged.

### D.2 Item T1 — G3 Class Imbalance: Terminal Bias Assessment

**Priority: ADVISORY**

In RUL regression tasks applied to the C-MAPSS dataset, the target variable is constructed under the piecewise-linear cap convention introduced by Saxena & Goebel (2008) [1]. The capped target is defined as:

$$\tilde{y}_i = \min(y_i, R_{\max})$$

where  $R_{\max}$  is typically set to 125 cycles. This convention creates a structural asymmetry in the label distribution: the early portion of each engine trajectory is assigned a constant ceiling label ( $\tilde{y}_i = 125$ ), whilst the terminal portion carries the only monotonically informative labels. The Critical band — engines with true RUL  $\leq 30$  cycles — is therefore substantially underrepresented relative to the cruise-phase majority class.

The G3 assessment quantifies terminal bias through the Critical-band mean residual, defined as the average signed prediction error across all engines whose true RUL falls within the Critical band:

$$\overline{\varepsilon}_{\text{crit}} = \frac{1}{|C|} \sum_{i \in C} (\hat{y}_i - y_i)$$

$$C = \{i : y_i \leq 30\} \quad (\text{Critical} - \text{band subset})$$

A positive signed value indicates systematic over-prediction (conservative bias), which is operationally tolerable under the NASA asymmetric scoring penalty structure. A negative value indicates under-prediction, which is operationally hazardous as it risks clearing engines for continued operation when they are near end-of-life.

---

**Critical-band mean residual = +2.33 cycles. Small positive bias confirmed. Directionally safe under NASA scoring convention. CONDITIONAL RESOLVED.**

---

The observed value of  $\overline{\varepsilon}_{\text{crit}} = +2.33$  cycles confirms that the model slightly over-predicts RUL in the terminal band. No resampling, focal weighting, or label smoothing was required for the  $v_0$  baseline.

### D.3 Item T2 — s16 Family Ablation via SHAP

**Priority: MANDATORY**

The s16 sensor family was constructed in Phase 3 as part of the kinematic feature expansion. SHAP (SHapley Additive exPlanations) TreeExplainer values Lundberg & Lee (2017) [5] were computed on a 5,000-row held-out validation sample to assess the marginal predictive contribution of each feature. The SHAP value for feature  $j$  on instance  $i$  is defined via the Shapley value from cooperative game theory [5]:

$$\varphi_j(i) = \sum_{S \subseteq \mathcal{F} \setminus \{j\}} \frac{|S|! \cdot (|\mathcal{F}| - |S| - 1)!}{|\mathcal{F}|!} [f(S \cup \{j\}) - f(S)]$$

where  $\mathcal{F}$  = full feature set,  $f(S)$  = model output conditioned on feature subset  $S$ .

The mean absolute SHAP value (MAS) across all instances provides a global attribution rank for each feature:

$$\text{MAS}_j = \frac{1}{N} \sum_{i=1}^N |\varphi_j(i)|$$

*SHAP asks: if I progressively add or remove this feature from the model's input, how much does the prediction change on average? Features that consistently shift the prediction substantially receive high attribution scores. Features that barely move the needle can be safely excluded without degrading model performance — particularly for neural networks, where low-signal features introduce numerical instability rather than predictive benefit.*

**s16 family aggregate SHAP attribution = 0.88% of total. Excluded from NN matrices. XGBoost matrices retain all 91 features. RESOLVED.**

The s16 family contributes 0.88% of total SHAP attribution across the 91-feature manifold. This falls below the exclusion threshold for neural architectures, where higher sensitivity to input scale and collinearity makes low-signal features a liability. The s16 family was excluded from `X_train_nn`, `X_val_nn`, and `X_test_nn`, reducing the NN feature count from 91 to 87. XGBoost matrices are unmodified.

### D.4 Item T3 — s9\_s14\_ratio: SHAP Audit & Percentile Clipping

**Priority: MANDATORY**

The ratio feature s9/s14 — constructed during Phase 3 kinematic expansion as a proxy for fan-speed-to-bypass-flow efficiency — was found to exhibit extreme distributional tails attributable to near-zero values in the s14 denominator under specific operating regimes. This constitutes a structural ratio degeneracy problem. The clipping operation applies percentile bounds as follows:

$$r_{\text{clipped}} = \text{clip}(r, P_1, P_{99})$$

where  $P_1$  and  $P_{99}$  are the 1st and 99th empirical percentiles of  $X_{\text{train}}$  only.

The training-set-only percentile constraint is mandatory: computing bounds from the full dataset (including validation or test rows) would constitute test-set leakage into the preprocessing step, violating the zero-leakage protocol.

**Table D.1.  $s9\_s14\_ratio$  Distributional Statistics: Raw vs. Post-Clip**

| Statistic                       | Raw Distribution | Post-Clip Distribution |
|---------------------------------|------------------|------------------------|
| Standard Deviation ( $\sigma$ ) | 650.60           | 2.71                   |
| Clip Bound — Lower ( $P_1$ )    | —                | -12.62                 |
| Clip Bound — Upper ( $P_{99}$ ) | —                | +14.74                 |
| Reduction Factor                | —                | ~240x                  |

**Post-clip sigma = 2.71 (raw sigma = 650.60). Reduction factor ~240x. SHAP confirms genuine**

*Imagine a ratio that occasionally explodes to a value of 50,000 because one number in the denominator dips close to zero momentarily. That single extreme value would dominate any gradient-based learning signal, making the feature essentially unusable and causing training instability. Clipping removes these pathological extremes by capping values at the 1st and 99th percentile boundaries — retaining 98% of the real distributional signal whilst eliminating outliers.*

**signal retained. Feature retained across all matrices. RESOLVED.**

## D.5 Item T4 — $s8\_s13\_ratio$ : SHAP Audit

**Priority: ADVISORY**

The ratio feature  $s8/s13$  — constructed as a secondary bypass efficiency indicator — was audited via SHAP to determine whether its inclusion provides independent predictive signal beyond the constituent raw sensors  $s8$  and  $s13$  individually. This distinguishes features that capture genuine interaction effects from those that merely duplicate information already present in the base sensor set. The relevant attribution statistic is:

$$MAS_{s8/s13} = 0.077 \quad (\text{rank 51 of 91 features})$$

*When you build a ratio from two existing measurements, you must verify that the ratio itself tells you something that neither measurement alone already captures. SHAP quantifies this: if the model assigns meaningful attribution to the ratio independently of the two raw sensors, the ratio is carrying genuine interaction information. Rank 51 of 91 confirms a mid-tier contribution — not a top driver, but not noise either.*

**MAS = 0.077, rank 51/91. Marginal additive contribution confirmed. Feature retained in all matrices. DOCUMENTED.**

## D.6 Item T5 — Yeo-Johnson Transformation on CPR and $E_{thermal}$

**Priority: MANDATORY**

Two virtual sensor features constructed in Phase 3 exhibit severe positive skewness in their raw distributions:

$$CPR = \frac{s_7}{s_3}$$

$$E_{thermal} = \frac{(s_{12} \times s_{11})}{s_9}$$

Skewness in input features is architecturally neutral for tree-based methods such as XGBoost, which operate via recursive binary splits and are invariant to monotone transformations. However, neural architectures with gradient-based optimisers are materially impacted: fat distributional tails inflate the effective learning-rate contribution of extreme values and destabilise batch normalisation statistics.

The Yeo-Johnson power transformation [14] handles both positive and negative values, making it appropriate for z-scored sensor data. For non-negative values, the transform reduces to:

$$\psi_{\lambda}(x) = \frac{(x+1)^{\lambda} - 1}{\lambda} \quad \text{for } \lambda \neq 0, \quad x \geq 0$$

$$\psi_{\lambda}(x) = \ln(x+1) \quad \text{for } \lambda = 0, \quad x \geq 0$$

The  $v_0$  implementation applied a `log1p + StandardScaler` cascade as a computationally tractable approximation. The scaler parameters (mean and standard deviation) are fitted exclusively on  $X_{\text{train}}$  and applied to  $X_{\text{val}}$  and  $X_{\text{test}}$  without refitting, preserving the zero-leakage constraint:

$$x' = \ln(1 + x)$$

$$x_{\text{scaled}} = \frac{x' - \mu_{x'}}{\sigma_{x'}}$$

$\mu_{x'}$  and  $\sigma_{x'}$  fitted on  $X_{\text{train}}$  only; applied to  $X_{\text{val}}$  and  $X_{\text{test}}$

**Table D.2. Yeo-Johnson Skewness Correction Results: CPR and  $E_{\text{thermal}}$**

| Feature                                 | Raw Skewness | Post-Transform Skewness |
|-----------------------------------------|--------------|-------------------------|
| CPR (s7 / s3)                           | 41.20        | -0.028                  |
| $E_{\text{thermal}}$ ((s12 x s11) / s9) | 171.82       | -0.011                  |

Assert:  $|S_{\text{post}}| < 2.0$  for all transformed features

Post-transform absolute skewness values of  $|S| < 0.05$  for both features satisfy this gate assertion with a margin of approximately 40x. Near-Gaussian symmetry is confirmed.

*If you plot CPR values on a histogram, the distribution has a very long tail extending to the right: most values cluster near zero, but occasional extreme values exist far to the right. Neural networks learn by comparing gradients across many training examples in a batch; if one example has a CPR value of 500 whilst most have values near 5, that single example dominates the gradient update and the network learns poorly. Log1p compression squashes the tail, making the distribution far more symmetric and giving the network a balanced view of all training examples.*

**Skewness: CPR 41.20 → -0.028;  $E_{\text{thermal}}$  171.82 → -0.011. Gate assertion  $|S| < 2.0$  satisfied. Separate NN matrices produced:  $X_{\text{train\_nn}}$ ,  $X_{\text{val\_nn}}$ ,  $X_{\text{test\_nn}}$ . XGBoost matrices unmodified. RESOLVED.**

## D.7 Item T6 — Inverse-Frequency Regime Weighting for Neural Networks

**Priority: MANDATORY**

The unified FD00u dataset is constructed across four operating condition regimes (derived from FD001–FD004). Within FD00u, regime membership is imbalanced: regime 0 accounts for 42.4% of all training rows, with minority regimes substantially underrepresented. Tree-based methods are structurally resilient to regime imbalance due to their non-parametric split criteria. Neural networks trained via mini-batch stochastic gradient descent are not resilient: if regime 0 appears approximately 3x more frequently than minority regimes, the network optimises preferentially on majority-regime loss.

The inverse-frequency sample weight for a training row belonging to regime  $k$  is defined as:

$$w_k = \frac{1}{N_k} \cdot \frac{N_{\text{total}}}{K}$$

where  $N_k$  = count of training rows in regime  $k$ ,  $K$  = number of regimes

$$\text{Normalisation: } \sum_k w_k \cdot N_k = N_{\text{total}}$$

**Table D.3. Inverse-Frequency Regime Weights Applied to Neural Network Training**

| Regime                  | Training Row % | Assigned Weight ( $w_k$ ) | Relative Emphasis |
|-------------------------|----------------|---------------------------|-------------------|
| Regime 0 (dominant)     | 42.4%          | 0.305                     | Down-weighted     |
| Minority regimes (mean) | ~19% each      | ~1.20                     | Up-weighted       |

**Regime 0 weight = 0.305. Minority regime weights ~1.20. Applied to NN training batches only. XGBoost unaffected. RESOLVED.**

## D.8 Item T7 — CPR Denominator Guard (s3 Degeneracy)

**Priority: ADVISORY**

The CPR (Compressor Pressure Ratio) feature is defined as the ratio of HPC exit total pressure ( $s_7$ , P30) to HPC inlet total pressure ( $s_3$ , P3 stage):

$$\text{CPR} = \frac{s_7}{s_3}$$

In the raw sensor domain, near-zero values of  $s_3$  under specific operating conditions can produce ratio degeneracy, yielding CPR values approaching infinity. This constitutes a structural numerical instability. In the z-scored input space used for  $v_0$ ,  $s_3$  is regime-normalised and centred; its minimum observed value does not approach zero in the C-MAPSS training distribution. The T5 resolution ( $\log_{1p}$  cascade) further suppresses CPR tail behaviour, providing indirect mitigation. The correct permanent engineering fix is a denominator floor guard:

$$\text{CPR} = \frac{s_7}{\max(s_3, \varepsilon)}$$

where  $\varepsilon$  is a small positive constant (e.g., 1e-6 in raw sensor units)



*Dividing by a number that occasionally drops close to zero is mathematically analogous to dividing by zero — the result blows up to a very large number. In the current pipeline, the data transformation in T5 largely prevents this from causing problems within the C-MAPSS operating envelope. However, the underlying structural vulnerability remains: if a future dataset (e.g., N-CMAPSS in FusionCore v<sub>2</sub>) contains operating conditions where s3 more aggressively approaches zero, CPR will again produce extreme values. A proper denominator guard is the correct permanent fix.*

**DEFERRED to v<sub>1</sub>. T5 cascade provides sufficient mitigation for v<sub>0</sub> within C-MAPSS envelope. Registered as carry-forward gap item for v<sub>1</sub>/v<sub>2</sub> resolution.**

## D.9 Item T8 — fillna(0) Convention at Trajectory Boundaries

**Priority: ADVISORY**

Rolling-window and cumulative features constructed in Phase 3 (e.g., s\_j\_rmean, s\_j\_cumfatigue) require a minimum number of historical cycles to compute a meaningful window value. At the start of each engine trajectory (cycle index = 1), the rolling window is not yet populated, and the feature value is mathematically undefined. The v<sub>0</sub> convention assigns NaN → 0 for these boundary rows.

The physical justification is twofold. For rolling mean features at cycle 1, no degradation history exists; assigning the baseline value of zero in z-scored space is equivalent to asserting the engine is operating at nominal condition, which is physically correct for a fresh trajectory. For cumulative fatigue features at cycle 1, a cumulative sum of zero is exact, not an approximation.

*Imagine computing a 10-day moving average of temperature starting from day 1 of your records. On day 1 itself, you have no prior days to average over. Filling with zero is equivalent to stating: "at the very first moment of this engine's recorded life, there is no degradation signal yet" — which is physically reasonable, since the engine is being assessed from the start of its recorded operational history.*

**Cycle-1 boundary rows = 0.44% of total training data. Zero-fill physically justified. DOCUMENTED.**

## D.10 Item T9 — Phase 3 Gate 2 Reporting Discrepancy

**Priority: ADVISORY**

During Phase 3 notebook execution, Gate 2 produced a displayed feature count inconsistent with the confirmed 91-feature specification established by the Phase 3 feature engineering register. This created a documentation ambiguity that, if unresolved, could invalidate the traceability chain between Phase 3 output and Phase 4 input. Three candidate root causes were evaluated:

| Candidate Cause          | Description                                                                         | Assessment                                             |
|--------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------|
| (a) Genuine feature loss | Four features failed a silent filter or assertion during Gate 2 processing          | Excluded — array shapes verified programmatically      |
| (b) Display artefact     | Notebook cell rendered a count from an intermediate variable, not the final matrix  | CONFIRMED ROOT CAUSE                                   |
| (c) Register error       | Phase 3 feature engineering register over-counted features relative to actual array | Excluded — register cross-checked against column names |

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**Confirmed display artefact. `x_train.shape[1] = 91` verified. Feature count correct. DOCUMENTED.**

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*If a quality control checkpoint at the end of a manufacturing stage says, "87 parts passed inspection" but the production log says, "91 parts were manufactured," one of three things has occurred: four parts failed inspection, the counter has a display error, or the log is wrong. Leaving this unresolved means you cannot certify how many parts entered the next stage. In data pipelines, such discrepancies must be traced to their root cause before downstream work is certified.*